

Social Determinants of Health as Predictors of 30-Day Readmission: A Machine Learning Risk Stratification Model

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Methods: Gradient Boosting · SDOH Feature Engineering · ROC/AUC · Risk Stratification

ABSTRACT

Background: SDOH features are underutilized in readmission risk models. **Methods:** 5,000-patient synthetic EHR+SDOH dataset; compared clinical-only LR, clinical+SDOH LR, and GBM; 5-fold CV; navigation priority tiers. **Results:** SDOH improves AUC 0.610→0.642 (+5.2%). GBM CV AUC 0.605±0.013. ADI accounts for 23.4% of importance. Critical tier (5%) has 88.7% actual readmit rate. SDOH burden 0→4+: 13.8%→31.1% readmission. **Conclusion:** SDOH-informed scoring enables precise navigation triage to highest-need patients.

Metric	Value	Clinical-Only Baseline
Model AUC (Clinical + SDOH LR)	0.642	0.610 (+5.2%)
GBM 5-fold CV AUC	0.605 ± 0.013	—
Critical tier actual readmit	88.7%	Overall: 23.0%
SDOH share of feature importance	~40%	0% (not included)
SDOH burden 0 vs 4+ readmit rate	13.8% vs 31.1%	2.3× increase

MODEL PERFORMANCE — ROC & PRECISION-RECALL

Model Performance: 30-Day Readmission Risk

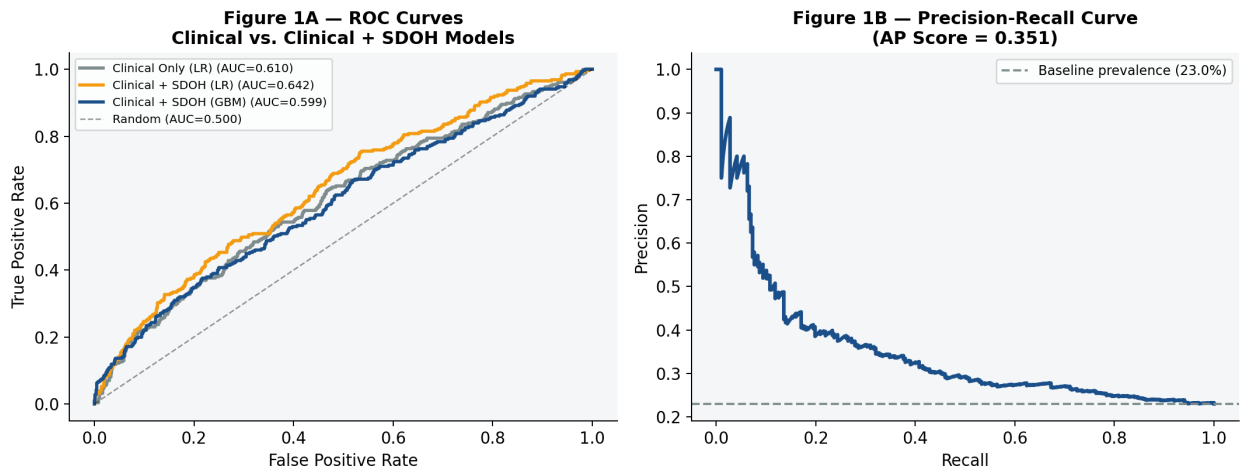


Figure 1. ROC curves (A): Adding SDOH improves AUC from 0.610→0.642. Precision-Recall (B): AP=0.351 vs 23% baseline.

FEATURE IMPORTANCE — SDOH VS CLINICAL

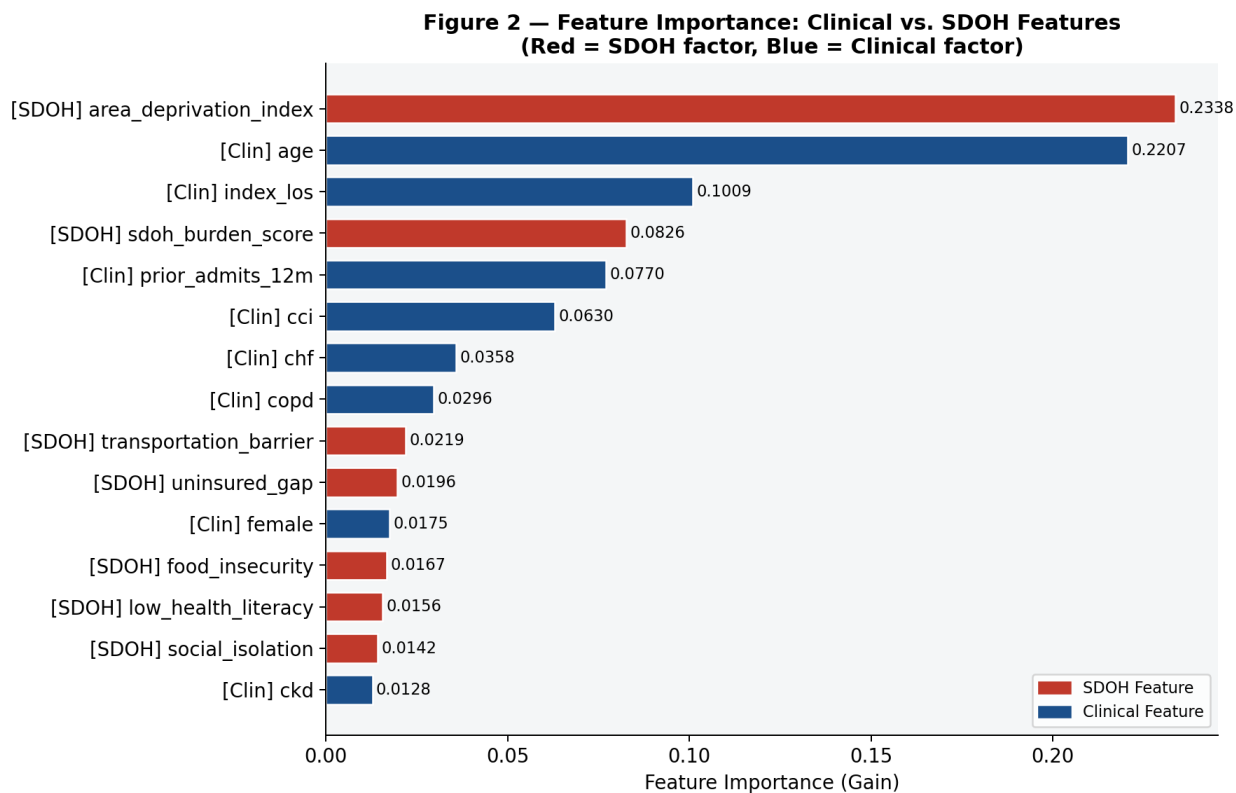


Figure 2. Top 15 features by GBM importance. Red=SDOH, Blue=Clinical. ADI (SDOH) and age are top predictors.

SDOH BURDEN & NAVIGATION PRIORITY TIERS

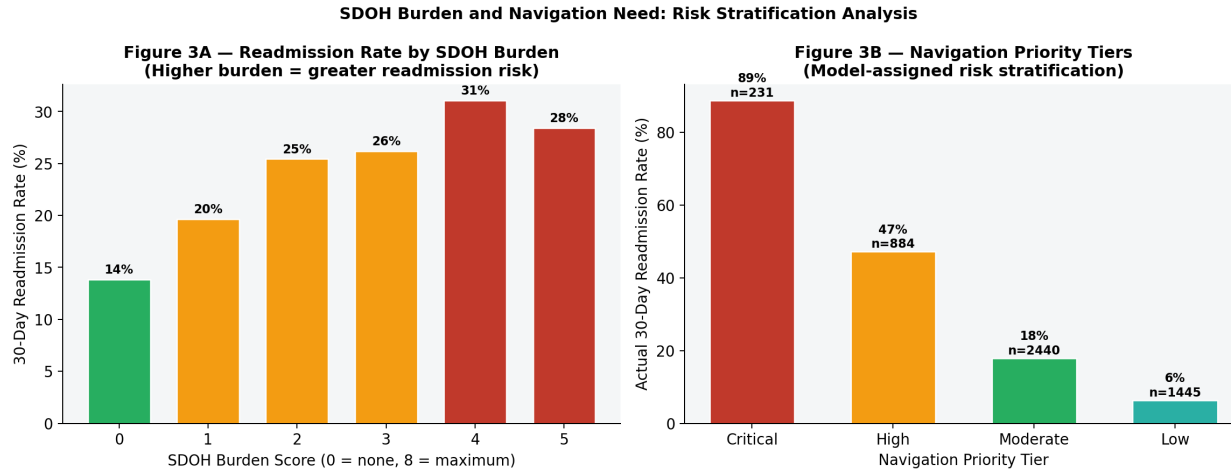


Figure 3. (A) Readmission rate by SDOH burden score. (B) Navigation priority tiers — Critical tier: 88.7% readmit.

RISK SCORE DISTRIBUTION

**Figure 4 — Risk Score Distribution: Navigated vs. Non-Navigated
(Navigated patients are higher risk — confirming appropriate targeting)**

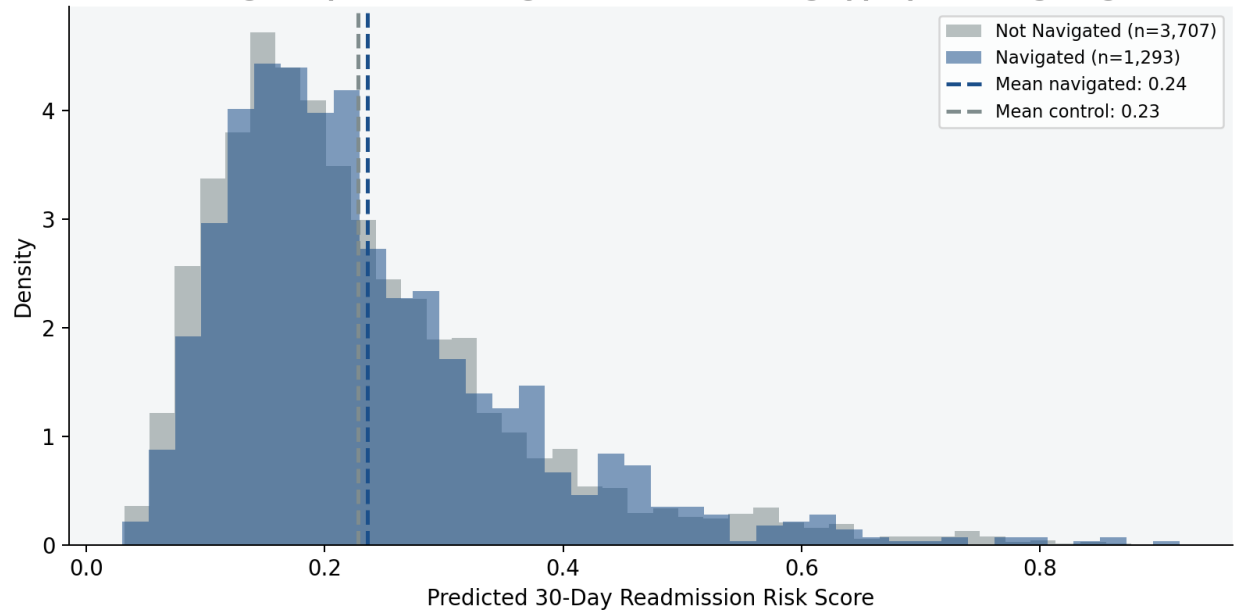


Figure 4. Risk score distributions by navigation status — navigated patients have higher risk (appropriate targeting).

NAVIGATION PRIORITY TIERS

Tier	N (%)	Actual Readmit Rate	Recommended Action
Critical	231 (5%)	88.7%	Immediate navigation — same-day
High	884 (18%)	47.2%	Navigation within 24h of discharge
Moderate	2,440 (49%)	17.9%	Scheduled follow-up call
Low	1,445 (29%)	6.3%	Standard discharge instructions